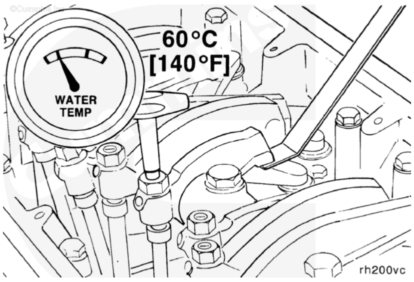
Trainings

Adjust

STC

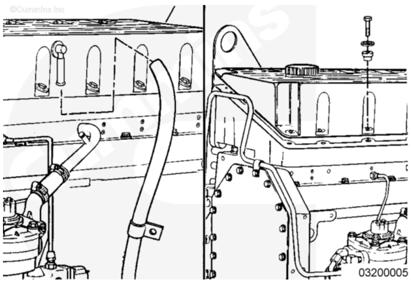


Valves and injectors **must** be correctly adjusted for the engine to operate efficiently. Valve and injector adjustment **must** be performed using the values listed in this section. The accompanying table gives the adjustment limits for STC engines.

Adjust the valves and the injectors at each 1,500 hour maintenance interval. If the valves and injectors have been adjusted during troubleshooting or before the 1,500 hour scheduled maintenance interval, adjustment is **not** required at this time.

Valve Lash Adjustment			
	mm		in
Intake Valve	0.35	MIN	0.014
Exhaust Valve	0.69	MIN	0.027

All overhead, valve and injector, adjustments **must** be made when the engine is cold, any stabilized coolant temperature at 60°C [140°F] or below.



Remove the crankcase breather tube from the crankcase breather outlet.

Remove the 16 capscrews, isolators, and spacers from the cover.

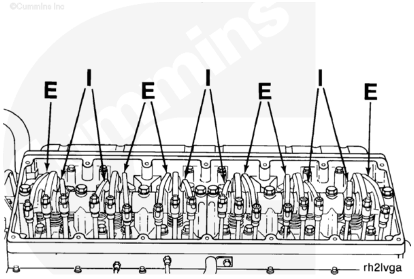
Remove the rocker lever cover and gasket.

Each cylinder has three rocker levers:

- The long rocker lever (E) is the exhaust lever.
- The center rocker lever is the injector lever.

- The short rocker lever (I) is the intake lever.

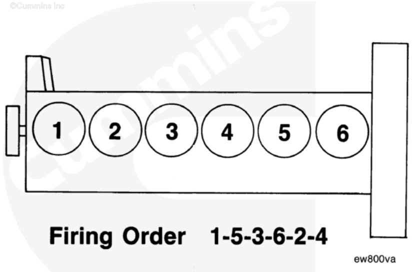
Reference the illustration for valve rocker lever locations.



The crankshaft rotation is **clockwise** when viewed from the front of the engine.

The cylinders are numbered from the front gear cover end of the engine.

The engine firing order is 1-5-3-6-2-4.



The valves and injectors on the same cylinders are **not** adjusted at the same index mark on the accessory drive pulley on STC engines.

One pair of valves and one injector are adjusted at each pulley index mark before rotating the accessory drive to the net index mark.

Two crankshaft revolutions are required to adjust all the valves and injectors.

STC
Injector and Valve Adjustment Sequence

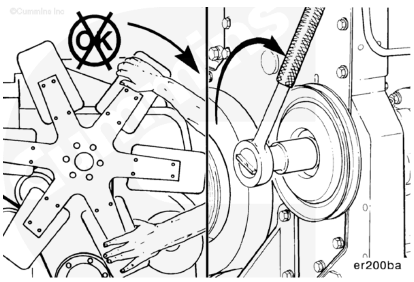
Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Injector	Set Cylinder Valve
Start	A	3	5
Advance to	B	6	3
Advance to	C	2	6
Advance to	A	4	2
Advance to	B	1	4
Advance to	C	5	1

Firing Order: 1-5-3-6-2-4

ol200vi

⚠ WARNING ⚠

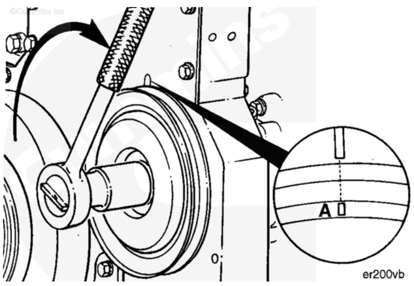
Do not straighten a bent fan blade or continue to use a damaged fan. A bent or damaged fan blade can fail during operation and cause personal injury or property damage.



The valve set marks are located on the accessory drive pulley. The marks align with a pointer on the gear cover.

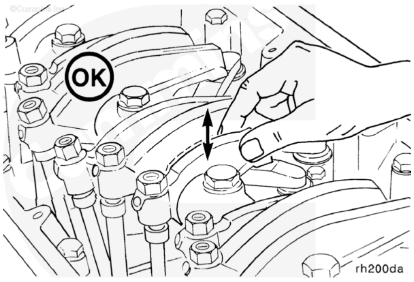
Use the accessory drive shaft to rotate the crankshaft.

The adjustment can begin on any valve set mark. In the following example the adjustment will begin on the “A” valve set mark with cylinder number five valves closed and cylinder number three injector ready for adjustment.



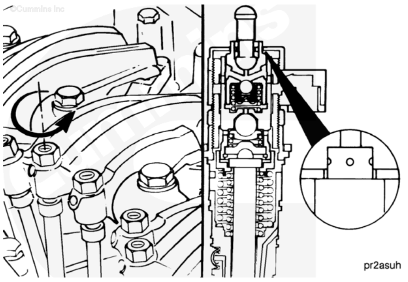
When the “A” mark is aligned with the pointer, the intake and exhaust valves for cylinder number five **must** be closed. If these conditions are **not** correct, cylinder number four injector and cylinder number two valves **must** be ready to set.

Both valves are closed when both rocker levers are loose and can be moved from side to side.



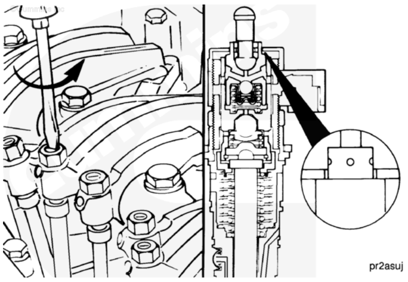
Loosen the injector adjusting screw locknut on cylinder number 3. Tighten the adjusting screw until all the clearance is removed from the injector train.

Tighten the adjusting screw one additional turn to seat the link correctly.

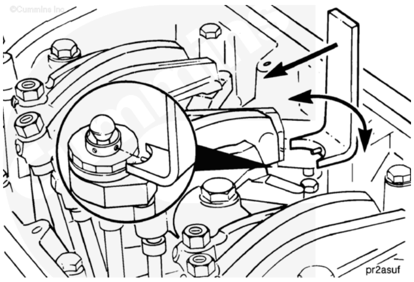


Loosen the injector adjusting screw until the STC tappet touches the top-cap of the injector.

Be sure to loosen the adjusting screw enough so there is no preload on the injector. This will be accomplished when the rocker lever is loose enough to move.

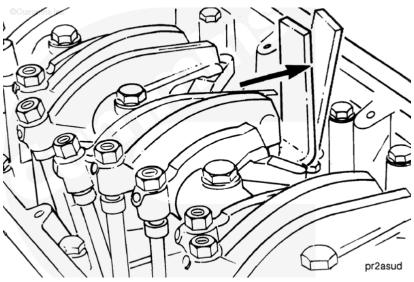


Place the STC tappet adjusting tool, Part Number 3823348, or equivalent, on the upper surface of the STC injector top-cap. Rotate the tool around the tappet until the tool's locating pin is inserted into one of the four holes in the top of the tappet.



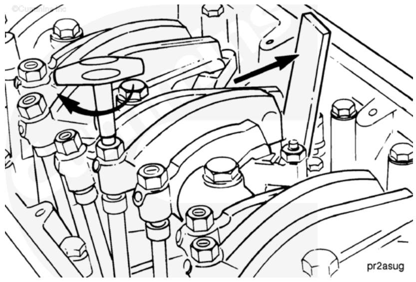
Note : Apply only enough force to the tool to hold the tappet in the maximum upward position. Excess force will cause the tool to break.

Apply thumb pressure to the tool handle to hold the tappet in the maximum upward position.



⚠ CAUTION ⚠

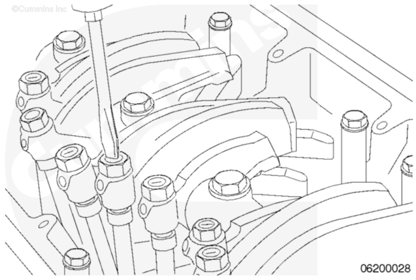
An overtightened setting on the injector adjusting screw will produce increased stress on the injector train and the camshaft injector lobe, which can result in engine damage.



Use torque wrench, Part Number 3376592, or equivalent, to tighten the adjusting screw while holding the tappet in the maximum upward position.

Torque Value: 0.7 n•m [6 in-lb]

Hold the adjusting screw in this position. The adjusting screw **must not** turn when the locknut is tightened.



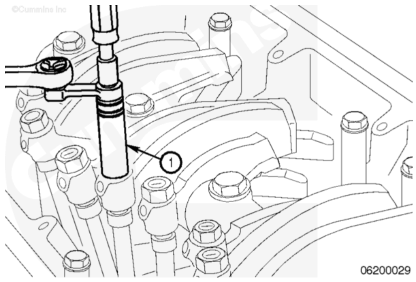
Tighten the adjusting screw.

- Without Torque Wrench Adapter:

Torque Value: 61 n•m [45 ft-lb]

- With Torque Wrench Adapter (1):

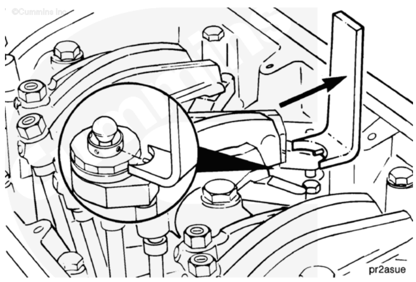
Torque Value: 47 n•m [35 ft-lb]



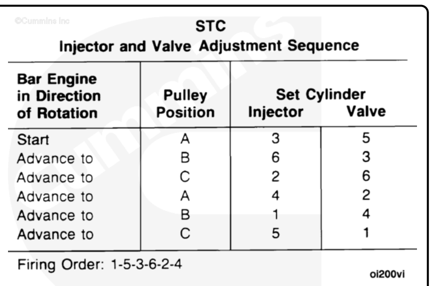
The tappet tool **must** be removed before rotating the crankshaft to prevent damage to the tappet.

Remove the tappet adjusting tool.

Check to make sure the injector push rod can be rotated by hand. If it can **not**, the setting is too tight.



Adjust the valves on the appropriate cylinder according to the sequence chart before rotating the accessory drive to the next valve set mark.



STC

Injector and Valve Adjustment Sequence

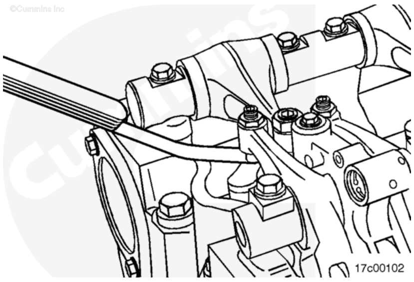
Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Injector	Set Cylinder Valve
Start	A	3	5
Advance to	B	6	3
Advance to	C	2	6
Advance to	A	4	2
Advance to	B	1	4
Advance to	C	5	1

Firing Order: 1-5-3-6-2-4

oi200vi

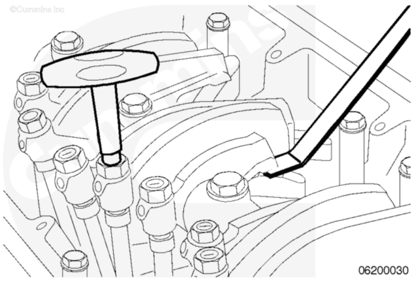
Select a feeler gauge for the correct valve lash specification.

Valve Lash Adjustment			
	mm		in
Intake Valve	0.35	MIN	0.014
Exhaust Valve	0.69	MIN	0.027



Insert the feeler gauge between the top of the crosshead and the rocker lever pad.

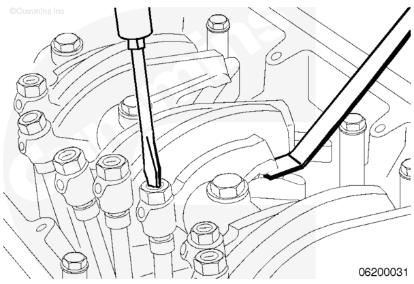
Two different methods for establishing valve lash clearance are described below. Either method can be used; however, the torque wrench method has proven to be the most concise. It eliminates the need to feel the drag on the feeler gauge.



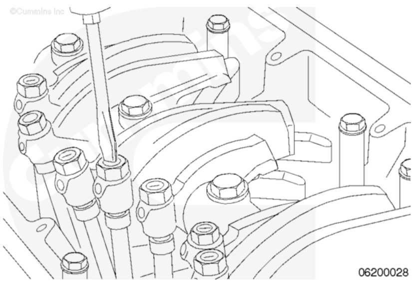
- Torque Wrench Method: Use an inch-pound torque wrench. Part Number 3376592, or equivalent, (normally used to set preload on top stop injectors), and tighten the adjusting screw.

Torque Value: 0.7 n•m [6 in-lb]

- Touch Method: Tighten the adjusting screw until a light drag is felt on the feeler gauge.



Hold the adjusting screw in this position. The adjusting screw **must not** turn when the locknut is tightened. Tighten the locknut.



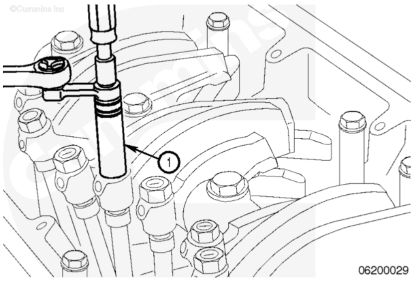
Tighten adjusting screw.

- Without Torque Wrench Adapter:

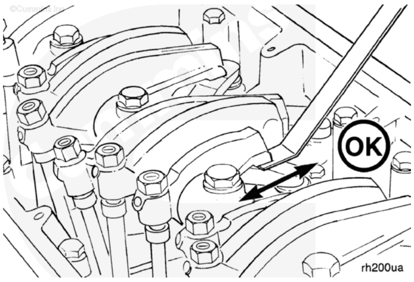
Torque Value: 61 n•m [45 ft-lb]

- With Torque Wrench Adapter (1):

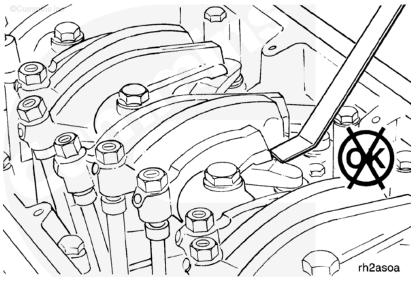
Torque Value: 47 n•m [35 ft-lb]



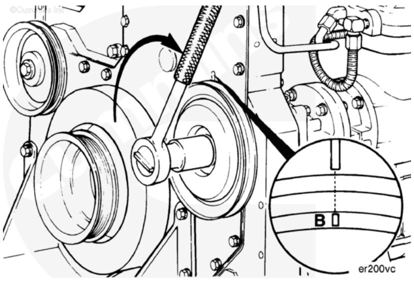
After tightening the locknut to the correct torque value, check to make sure the feeler gauge will slide backward and forward between the crosshead and the rocker lever with **only** a slight drag.



If using the touch method, attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker between the crosshead and the rocker lever pad. The valve lash is **not** correct when a thicker feeler gauge will fit.



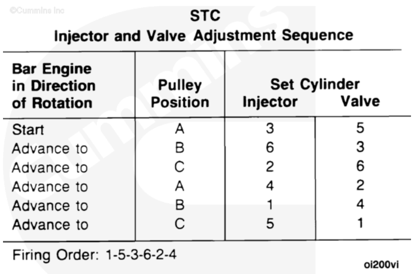
After adjusting the valves, rotate the accessory drive and align the next valve set mark on the accessory drive pulley with the pointer on the gear cover.



Adjust the appropriate injector and valves following the Injector and Valve Adjustment Sequence Chart.

Repeat the process to adjust all injectors and valves.

After adjusting all the injectors and valves, check the torque on the adjusting screw locknuts to make sure none were overlooked.


STC
Injector and Valve Adjustment Sequence

Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Injector	Set Cylinder Valve
Start	A	3	5
Advance to	B	6	3
Advance to	C	2	6
Advance to	A	4	2
Advance to	B	1	4
Advance to	C	5	1

Firing Order: 1-5-3-6-2-4

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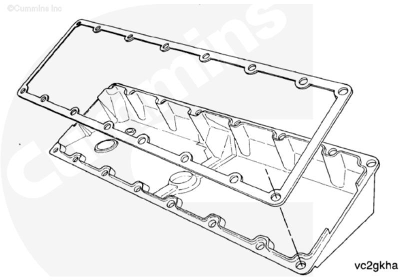
If the valve cover gasket is **not** damaged, it can be used again. If the gasket is damaged, it **must** be discarded and a new one used.

Install the gasket on the cover.

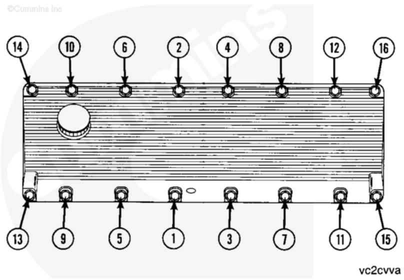


Install the cover on the rocker lever housing.

Install the 16 isolators, spacers, and capscrews in the cover.

Tighten the capscrews in the sequence shown.

Torque Value: 15 n•m [130 in-lb]



CELECT™ or CELECT™ Plus

Valves, injectors, and engine brakes **must** be correctly adjusted for the engine to operate efficiently. Valve, injector, and engine brake adjustment **must** be performed using the values listed in this section. The accompanying table gives the adjustment limits for CELECT™ and CELECT™ Plus engines.

Adjust the valves and the injectors at each 1,500 hours or 1 year maintenance interval. If the valves and injectors have been adjusted during troubleshooting or before the scheduled interval, adjustment is **not** required at this time.

CELECT™ Plus

Valve, Injector, and Brake Adjustment Specifications

	mm	in
Intake Valve	0.35	0.014
Exhaust Valve	0.68	0.027
Engine Brake	0.38	0.015

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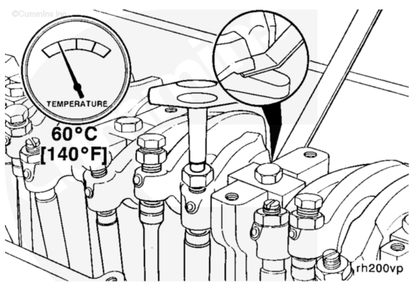
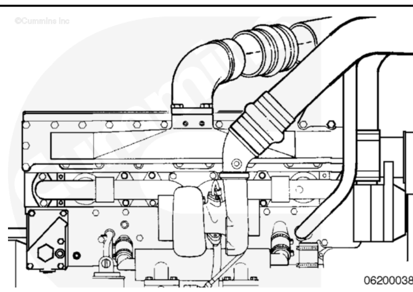
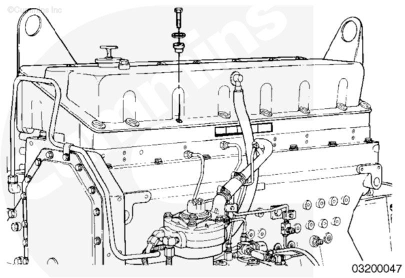
All valve and injector adjustments **must** be made when the engine is cold, any stabilized coolant temperature at 60° C [140°F] or below.



Remove the air piping from the intake manifold.

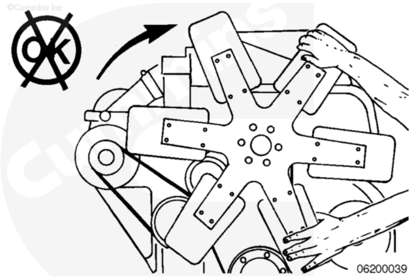


Remove the rocker lever cover.



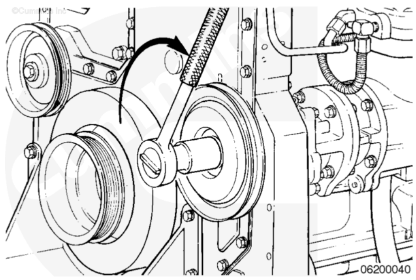
⚠ WARNING ⚠

Do not straighten a bent fan blade or continue to use a damaged fan. A bent or damaged fan blade can fail during operation and cause personal injury or property damage.



The valve set marks are located on the accessory drive pulley. The marks align with a pointer on the gear cover.

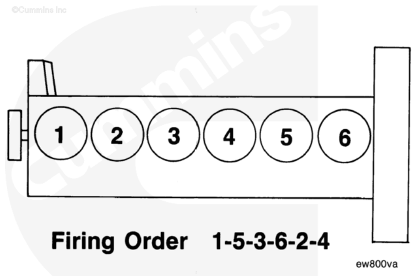
Use the accessory driveshaft to rotate the crankshaft.



The crankshaft rotation is **clockwise** when viewed from the front of the engine.

The cylinders are numbered from the front gear housing end of the engine.

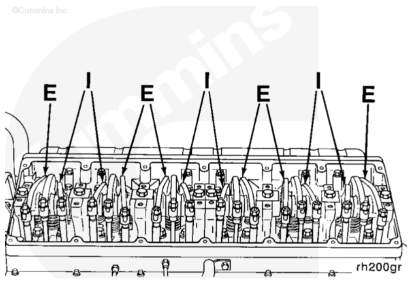
The engine firing order is 1-5-3-6-2-4.



Each cylinder has three rocker levers:

- The long rocker lever (E) is the exhaust lever.
- The center rocker lever is the injector lever.
- The short rocker lever (I) is the intake lever.

Reference the accompanying chart for valve rocker lever locations.



The valves and injectors on the same cylinders are adjusted at the same index mark on the accessory drive pulley on CELECT™ engines.

One pair of valves and one injector are adjusted at each pulley index mark before rotating the accessory drive to the next index mark.

Two crankshaft revolutions are required to adjust all the valves and injectors.

CELECT™
Injector and Valve Measurement Sequence

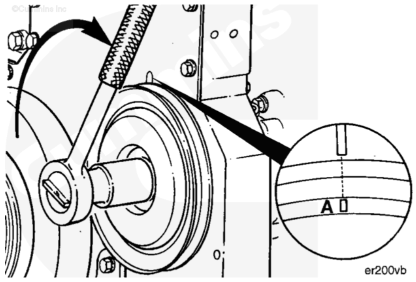
Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Injector	Set Cylinder Valve
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4

Firing Order: 1-5-3-6-2-4

oi200vh

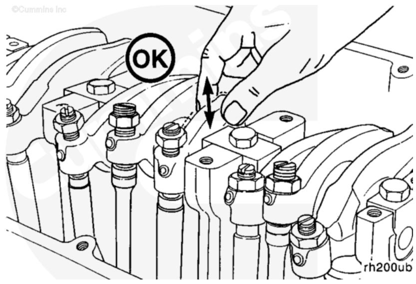
The adjustment can begin on any valve set mark. In the following example, the adjustment will begin on the “A” valve set mark with cylinder number 1 valves closed and ready for adjustment.

Rotate the accessory drive **clockwise** until the “A” valve set mark on the accessory drive pulley is aligned with the pointer on the gear cover.



When the “A” mark is aligned with the pointer, the intake and exhaust valves for cylinder number 1 **must** be closed. If these conditions are **not** correct, cylinder number six injector and valves **must** be ready to set. Set the injector and valves on the cylinder that both the intake and exhaust valve rocker lever arms are loose and can be moved from side to side.

Both valves are closed when both rocker levers are loose and can be moved from side to side.



Loosen the injector adjusting screw locknut.

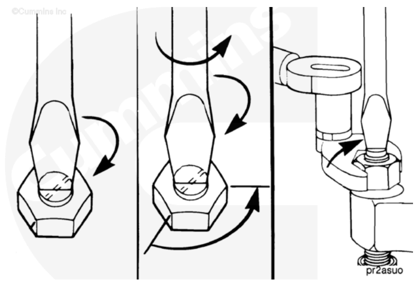
Use a screw driver or a box end wrench to adjust the screw. Bottom the injector plunger three or four times to remove the fuel.

Turn the adjusting screw in until it just bottoms the plunger.

Note : Do not use excess force when bottoming the plunger.

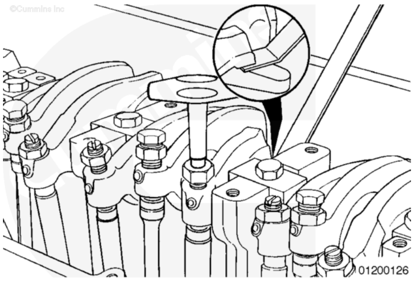
Back out the adjusting screw two flats, 120 degrees.

Hold the adjusting screw and tighten the locknut.

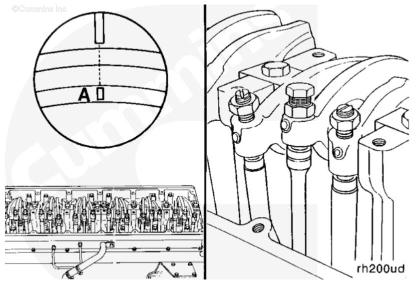


Torque Value: 61 n•m [45 ft-lb]

After setting the injector, set the valves on the same cylinder.

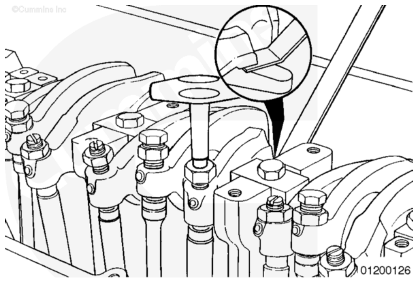


With the “A” valve set mark aligned with the pointer on the gear cover and both valves closed on the cylinder to be adjusted, loosen the adjusting screw locknuts on the intake and exhaust valves.



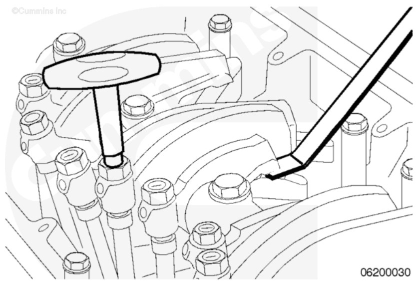
Select a feeler gauge for the correct valve lash specification.

Valve Lash Specification			
	mm		in
Intake	0.36	MIN	0.014
Exhaust	0.69	MIN	0.027



Insert the feeler gauge between the top of the crosshead and rocker lever pad.

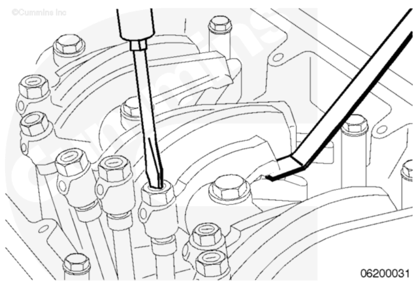
Two different methods for establishing valve lash clearance are described below. Either method can be used; however, the torque wrench method has proven to be the most consistent. It eliminates the need to feel the drag on the feeler gauge.



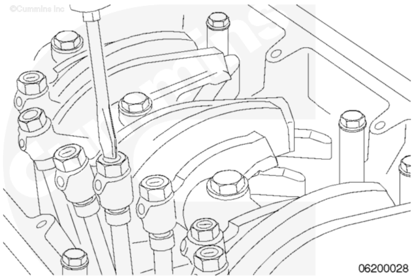
- Torque Wrench Method: Use the inch-pound torque wrench. Part Number 3376592, or equivalent (normally used to set preload on top stop injectors), and tighten the adjusting screw.

Torque Value: 0.7 n•m [6 in-lb]

- Touch Method: Tighten the adjusting screw until a light drag is felt on the feeler gauge.



Hold the adjusting screw in this position. The adjusting screw **must not** turn when the locknut is tightened. Tighten the locknut.



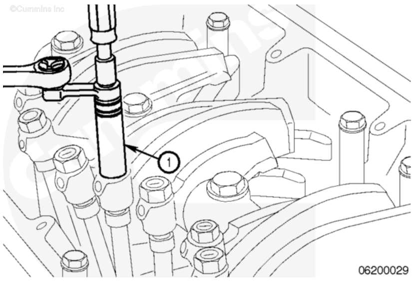
Tighten adjusting screw.

- Without Torque Wrench Adapter:

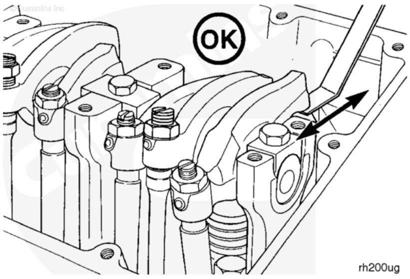
Torque Value: 61 n•m [45 ft-lb]

- With Torque Wrench Adapter (1):

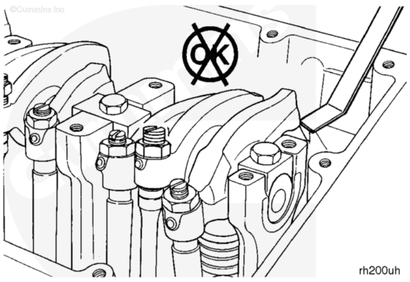
Torque Value: 47 n•m [35 ft-lb]



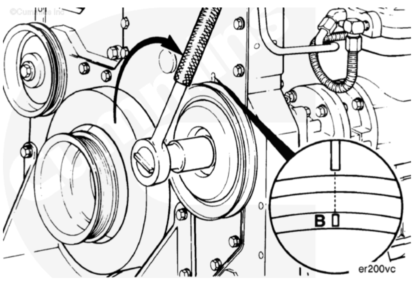
After tightening the locknut to the correct torque value, check to make sure the feeler gauge will slide backward and forward between the crosshead and the rocker lever with **only** a slight drag.



If using the touch method, attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker between the crosshead and the rocker lever pad. The valve lash is **not** correct when a thicker feeler gauge will fit.



After adjusting the injector and valves on the appropriate cylinder, rotate the accessory drive pulley and align the next valve set mark with the pointer on the gear cover.



Adjust the appropriate injector and valves following the injector and valve adjustment sequence chart.

Repeat the process to adjust all injectors and valves correctly.

CELECT™
Injector and Valve Measurement Sequence

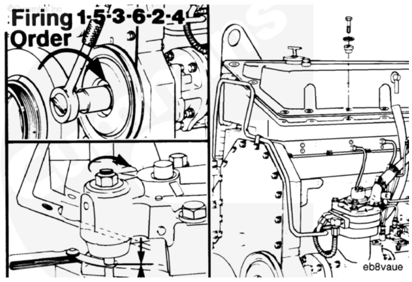
Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Injector	Set Cylinder Valve
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4

Firing Order: 1-5-3-6-2-4

oi200vh

If the engine is equipped with an engine brake, adjust the engine brake.

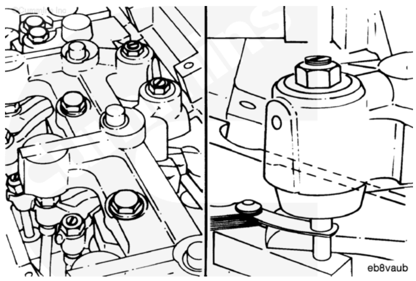
If the engine is **not** equipped with an engine brake, install the rocker lever cover.



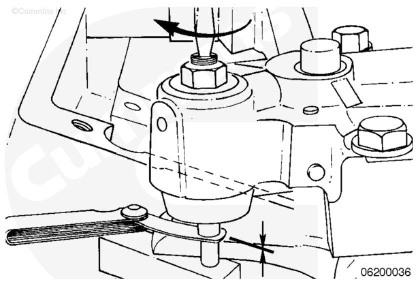
With Engine Brakes

Note : To obtain maximum brake operating efficiency and prevent engine damage by piston-to-valve contact, complete the following instructions carefully.

After adjusting the exhaust valves on the appropriate cylinder, insert a feeler gauge 0.38 mm [0.015 in] between the slave piston and the actuating pin in the crosshead.



Turn the slave piston adjusting screw down until it touches the feeler gauge.



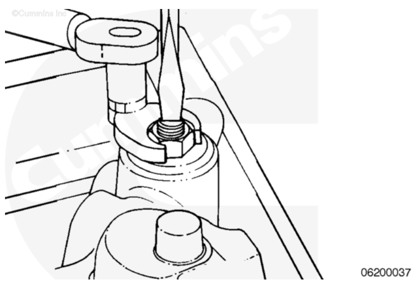
Hold the screw in position, and tighten the locknut.

- Without torque wrench adapter

Torque Value: 34 n•m [25 ft-lb]

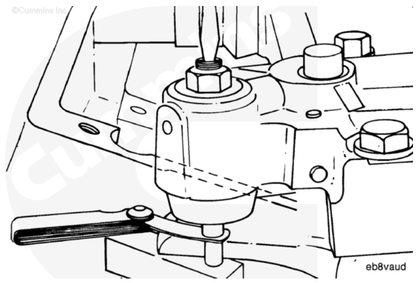
- With torque wrench adapter, Part Number 3163196, or equivalent:

Torque Value: 30 n•m [22 ft-lb]



⚠ CAUTION ⚠

After the slave piston adjusting screw locknut is tightened to the correct torque value, check the clearance with the feeler gauge again. Do not tighten the adjusting screws too tightly. The engine can be damaged.



Adjust the appropriate injector, valves, and engine brake following the Injector and Valve Adjustment Sequence Chart.

Repeat the process to adjust all injectors and valves.

After adjusting all the valves, injectors, and brakes, check the torque on the adjusting screw locknuts to make sure none were overlooked.

CELECT™
Injector and Valve Measurement Sequence

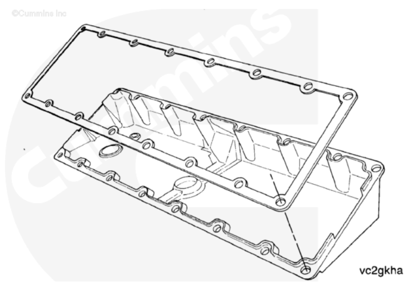
Bar Engine in Direction of Rotation	Pulley Position	Set Cylinder Injector	Set Cylinder Valve
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4

Firing Order: 1-5-3-6-2-4

oi200vh

If the valve cover gasket is **not** damaged, it can be used again. If the gasket was damaged, it **must** be discarded and a new one used.

Install the rocker lever cover gasket on the cover.

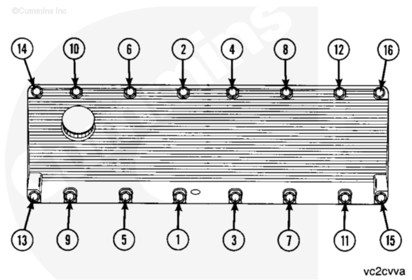


Install the cover on the rocker lever housing.

Install the 16 isolators, spacers, and capscrews in the cover.

Tighten the capscrews in the sequence shown.

Torque Value: 15 n•m [130 in-lb]



Last Modified: 07-Oct-2011